

Underbranch LaTeX Editor

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Background: LaTeX is the most commonly used typesetting language worldwide. It is built upon the TeX engine, and is the industry standard for research in math, physics, and computer science. However, mathematicians aren't always at a computer when they need to communicate an idea. Alongside several other students in COSC340: Software Engineering, I developed an online collaborative LaTeX editor and compiler called Underbranch. I propose a two-phase extension to this previous project. The first phase involves the team adding features and a friendly interface to Underbranch Web. The second phase is a mobile messaging app that supports LaTeX compilation and rendering in-chat, with a user-friendly experience that lends itself to fast input. Potential users include researchers, math students, and TAs who discuss math informally and currently are forced to resort to either pictures or plaintext. We aim to extend Underbranch to a messaging app through an extension in iMessage that enables users to view equations in chat. Users without the app should still be able to view equations in chat. For this writeup, I'll focus mostly on the mobile application, as extending the web interface should be relatively self-explanatory.

Project Needs for app:

1. Create a mobile app that provides an extension within iMessage to create embedded equations within messages.
2. Develop compilation backend; potentially TeXLive, KaTeX, SwiftMath, iosMath
3. Preview for users as they compose their equation; most likely some port of MathLive.
4. User accounts and authentication, mixed content messages (text and rendered LaTeX).

Project Needs for web:

1. Extend Underbranch Web with:
 - a. compilation endpoint
 - b. equation preview
 - c. improved operational transform for collaboration
 - d. common packages
 - e. project storage
 - f. potentially cloud compilation.

Project Scope for app:

1. This will involve building a Swift iMessage app with a custom math keyboard and a live preview.
2. We'd also like native LaTeX-to-image rendering.
3. Reach Goals: Another nice feature would be the ability to retroactively edit an equation that you'd previously sent, similar to the recent iMessage feature that allows users to edit their messages. A "saved-equation" library would also be a great quality-of-life addition.
4. We are explicitly NOT supporting Android. This is a Swift app and will remain on Apple alone.

Potential Technical Considerations:

1. LaTeX compilers already exist out there - there's no point in re-inventing the wheel. However, we have to consider that we'll have to embed PDFs or vector graphics within a widget that iMessage

will have to receive. I haven't looked deeply into this yet, but can foresee that this may be a difficult task.

2. None of the former Underbranch team members have used Swift before. We'll have to be sure to familiarize ourselves with the language before making any huge design decisions.
3. Rendering: We'd prefer a native LaTeX renderer like SwiftMath or iosMath over a non-native solution like KaTeX or MathJax. This will reduce memory usage.
4. I've done a little reading on these extensions, and they can be forced to run under a lower memory ceiling than full apps. We'll have to keep this in mind, especially when thinking about rendering.

Deliverables

1. The Messages extension is installable via TestFlight.
2. Custom math keyboard to increase speed of input.
3. Design documentation.
4. Usability evaluation report.

Assumptions and Dependencies:

1. Main assumption that we must satisfy: a non-user of the app must be able to see the rendered image.
2. Assumption: Both users are on iMessage on Apple products that explicitly support iMessage.
3. Assumption: The team can acquire Swift skills.
4. Dependency: Apple Developer Program (\$99/year) required to build the extension onto devices and distribute via TestFlight.
5. Dependency: Two physical iOS devices.